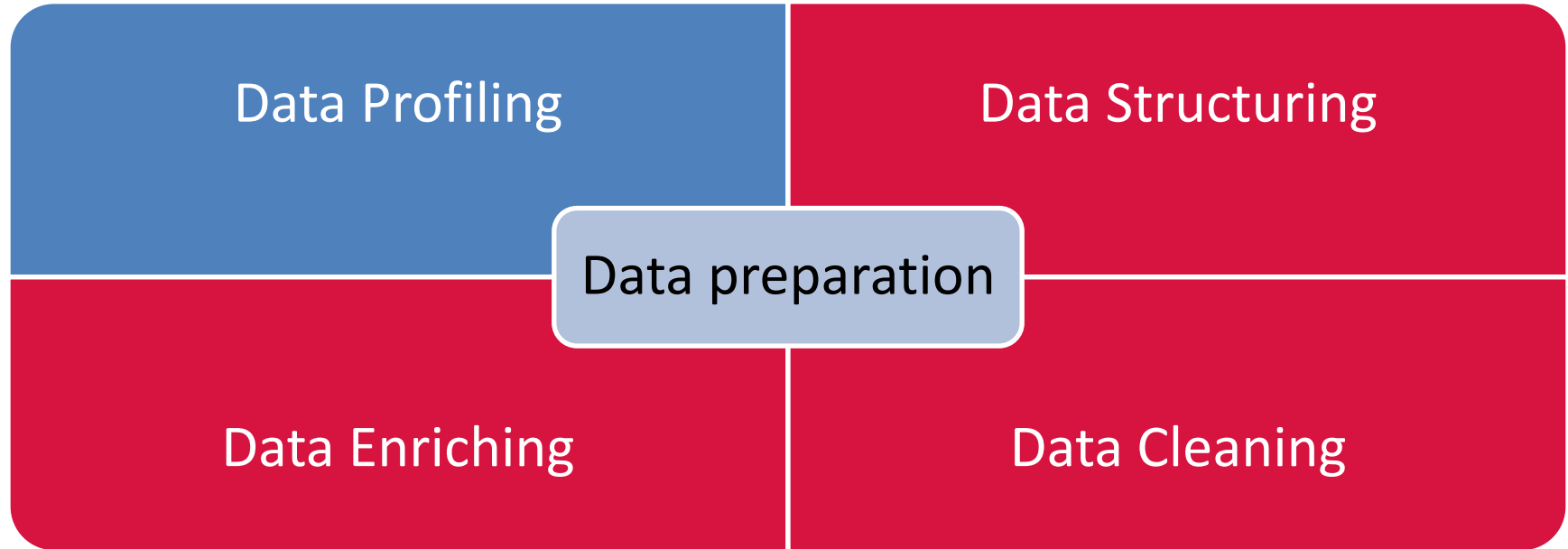


Data Science Fundamentals [SC3021]

Chapter 4: Data Preparation

Assoc. Prof. Melanie Herschel | CCDS

Data transformation tasks

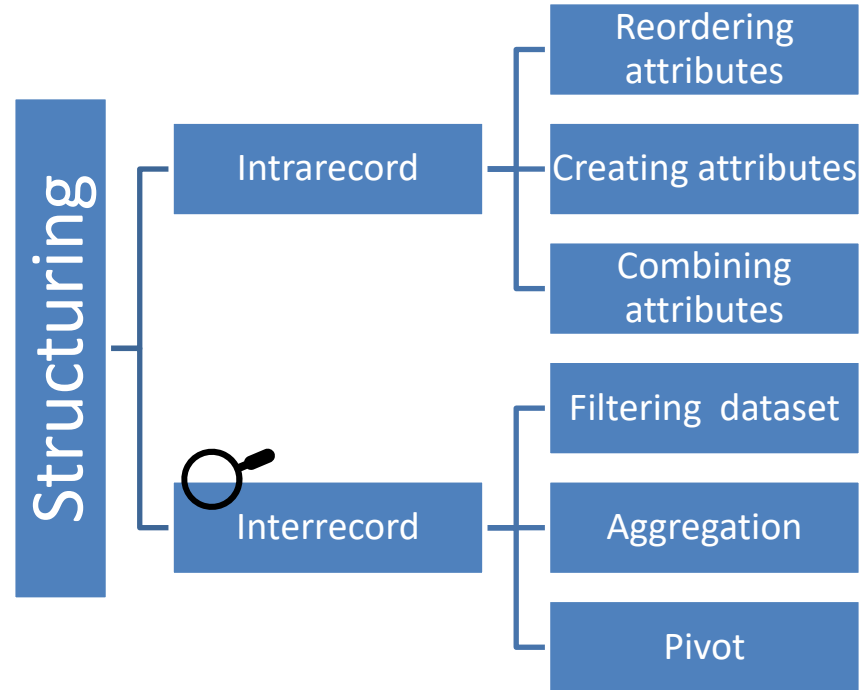




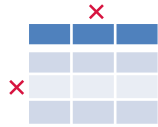
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Chapter 4.3: Data Transformation

Overview of structuring transformations



Interrecord structuring



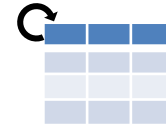
Filtering

Removal of columns or rows.



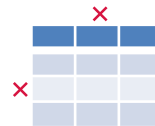
Aggregation

Attributes are added with values **extracted** from existing attributes



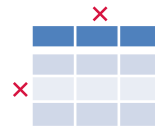
Pivot

Attributes are **merged** into a single attribute



Filtering

- **Removes** records or attributes from a dataset.
- Is considered a structuring operation when it is used to alter the **granularity** of a dataset by changing the types of represented records and attributes.
- Two **sub-types**:
 - **Record-based** filtering removes records (rows)
 - **Attribute-based** filtering removes attributes (columns)



Filtering - Examples

- **Record-based filtering:**
“Zoom-in” to only one type of entity that made each donation by limiting dataset to rows with Column7 = “PAC”
- **Attribute-based filtering:**
“Zoom-in” by retaining only relevant attributes.
- Both result in a finer-grained granularity than the original dataset.

Record-based

Column1	Column2	Column3	Column4
C00495580	N	YE	P
C00495580	N	YE	P
C00495580	N	YE	P
C00495580	N	YE	P
C00495580	N	YE	P
C00515049	N	YE	P
C00563601	N	YE	P
C00587030	N	YE	P
C00603084	N	YE	P
C00578997	A	YE	P
C00114439	A	YE	P
C00114439	A	YE	P
C00536664	A	YE	P
C00626390	A	YE	P

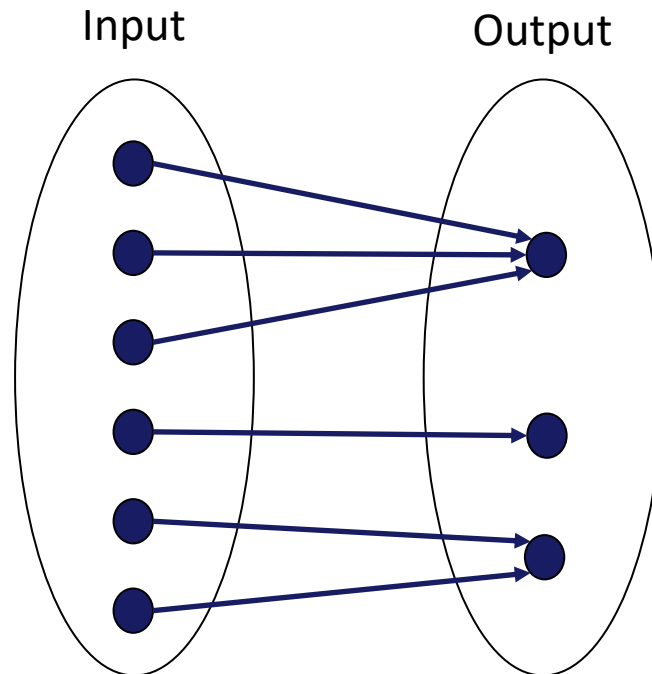
Attribute-based

Column1	Column2	Column3	Column4
C00276311	15	IND	NE
C0049535	15E	IND	VA
C0049535	15E	IND	NY
C0049535	15E	IND	DC
C0049535	15E	IND	DC
C0049535	15E	IND	VA
C0049535	15E	IND	VA
C0049535	15E	IND	VA
C0049535	15E	IND	CA
C0049535	15E	IND	NY
C0049535	15E	IND	MA
C0049535	15E	IND	VA
C0049535	15E	IND	FL
C0049535	15E	IND	VA
C0049535	15E	IND	VA
C0049535	15E	IND	VA
C0049535	15	IND	MA

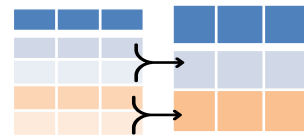


Aggregation

- **Combination** of data that obtains new attributes with a **shifted granularity** compared to the original data.
- Each **input record** maps to exactly one output record.
- Each **output record** combines one or more input records.
- Values of the new attributes in the output are obtained through **aggregation functions** (sum, mean, min, list concatenation, etc.) of input record attributes.



Aggregation – Example



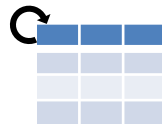
- Manipulate the granularity of the dataset so that each row summarizes the campaign contributions made to each campaign committee.
- For each campaign committee, we create three new columns:
 - The average contribution made to it
 - The sum of contributions made to it
 - The number of contributions made to it

```
# Group by 'Column1' and calculate the average, sum, and count of 'Column15'
grouped = df.groupby('Column1')['Column15'].agg(['mean', 'sum', 'count'])

# Rename the columns for clarity
grouped = grouped.rename(columns={
    'mean': 'Average_Column15',
    'sum': 'Sum_Column15',
    'count': 'Count_Column15'
})

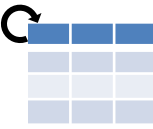
print(grouped)
```

Column1	Average_Column15	Sum_Column15	Count_Column15
C00001016	101.250000	86670	856
C00001198	896.275862	25992	29
C00001214	1151.478261	26484	23
C00001727	500.000000	500	1
C00002469	92.820724	112870	1216
...	...	...	...
C00629956	2000.000000	2000	1
C00630012	5000.000000	5000	1
C00630103	500.000000	500	1
C00630186	328.000000	1640	5
C90016742	29.500000	59	2



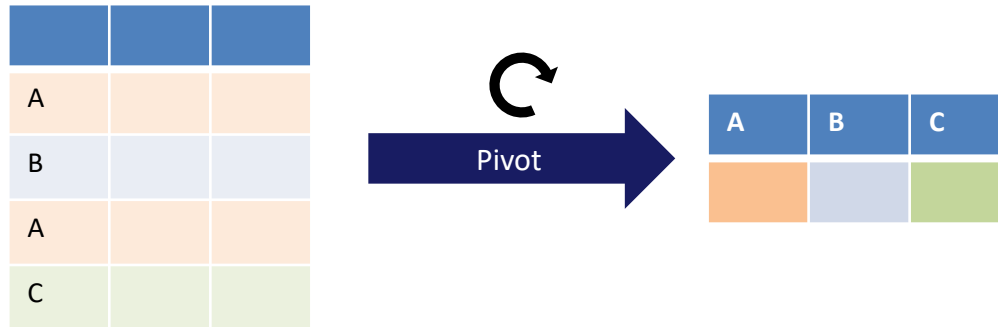
Pivot

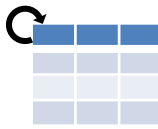
- Flips between row values and columns
- Two types of pivot:
 - Row-to-column pivot
 - Column-to-row pivot (aka unpivot)



Row-to-column pivot

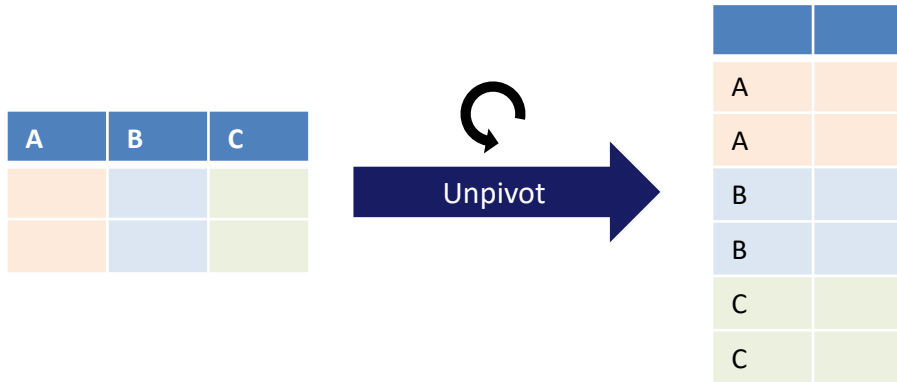
- Output records are based on multiple input records
- Values of output record attributes are based on some aggregation of input record attribute values.
- Frequently combined with aggregation, such that input records can contribute to multiple output records.





Column-to-row pivot (unpivot)

- Each input record maps to multiple output records.
- Each output record is based on exactly one input record.
- Values of separate attributes of the input are unified into a single attribute in the output.



Pivot – Example

- Six distinct values of Column7 identify different source types of contributions.
- Contributions made from various US states.
- Aggregation + pivot to count the number of contributions per state (i.e., Column10 remains) and source type (pivots from row values to columns)

```
#Create data grouped by source type and state
grouped = df.groupby(
    ['Column7', 'Column10'])['Column1'].agg(['count'])

#Pivot grouped to get Column7 as columns
pivot_table = grouped.pivot_table(
    index='Column10',
    columns='Column7',
    values='count',
    aggfunc='sum',
    fill_value=0
)

print(pivot_table)
```

Column7 Column10	CAN	CCM	COM	IND	ORG	PAC
AA	0	0	0	1	0	0
AE	0	0	0	2	0	0
AK	0	0	0	402	0	0
AL	0	0	0	2242	1	0
AR	0	0	0	1683	0	0
AZ	0	0	0	2278	0	0
CA	2	0	0	10301	12	0

Unpivot – Example

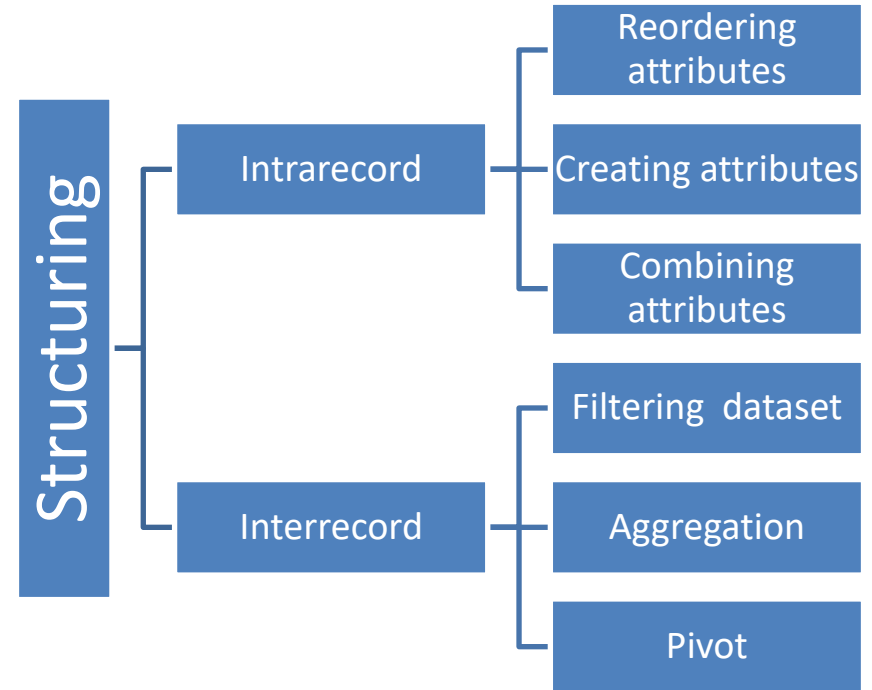
- “Reverse” the previous example.
- Each state will be associated with six values that correspond to the six attributes we will “put back” from the column header to the data records.

```
# Unpivot the pivot table
unpivoted_df = pd.melt(
    pivot_table.reset_index(),
    id_vars='Column10',
    value_vars=['CAN', 'CCM', 'COM', 'IND', 'ORG', 'PAC'])
print(unpivoted_df.sort_values(by='Column10'))
```

	Column10	Column7	value
0	AA	CAN	0
57	AA	CCM	0
171	AA	IND	1
228	AA	ORG	0
285	AA	PAC	0
..	...	...	...
227	ZZ	IND	28
113	ZZ	CCM	0
56	ZZ	CAN	0
284	ZZ	ORG	0
341	ZZ	PAC	0

Summary of structuring transformations

- Structuring as a transformation action involves **changing the structure and granularity** of a dataset.
- Structuring transformations create **new data based on data already present** in the dataset.



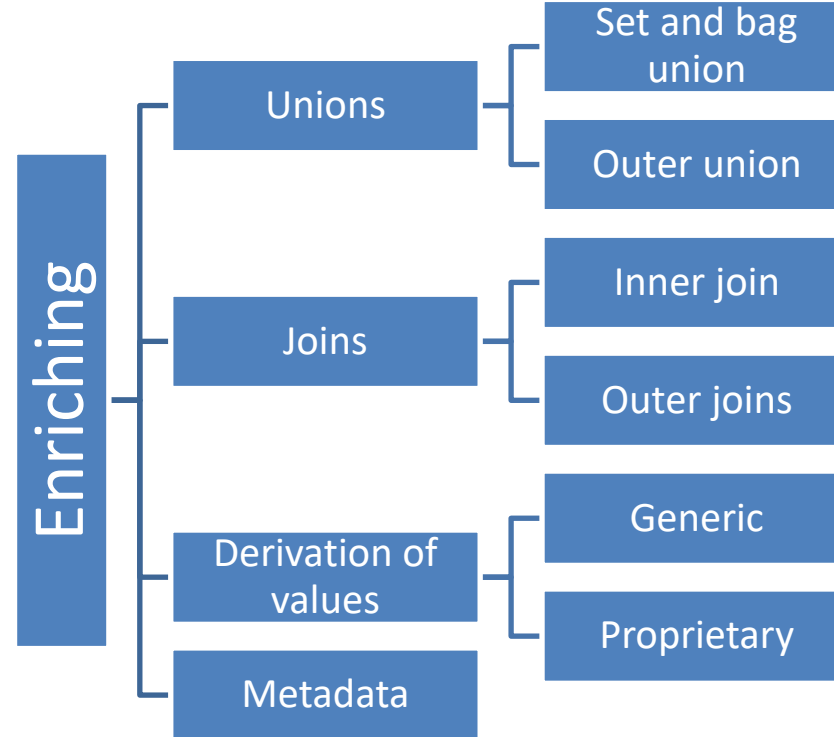
ENRICHING



Overview of enriching

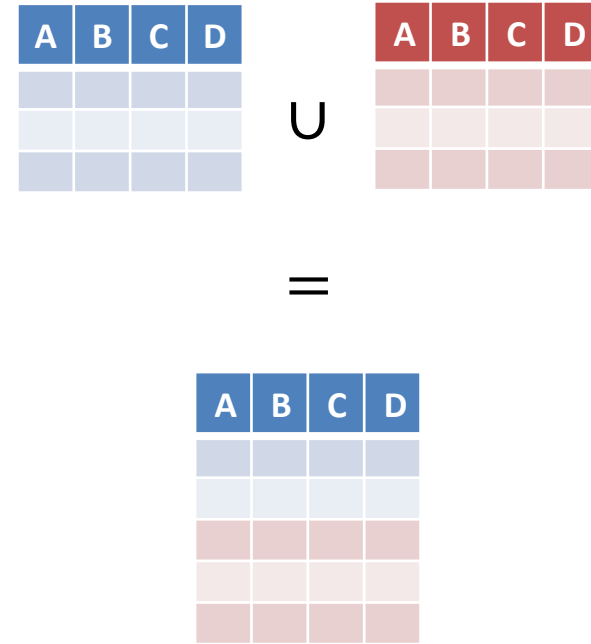
- Results in the net **addition of information** to a dataset.
 - Inserts additional records or attributes from other related datasets.
 - Use formulas to calculate new fields.
- Three **primary types** of enriching transformations:
 - Unions
 - Joins
 - Deriving new fields

Overview of enriching



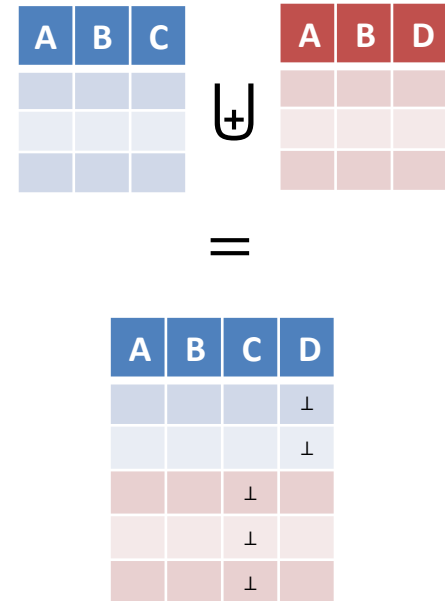
Unions – Set and bag union

- Given relations R and S with compatible schemas, **union** ($\cup$) creates a relation that includes **all tuples from R and all tuples from S** .
- Compatible schemas** have the same number of attributes and each attribute $R.A_i$ has the same data type as $S.A_i$.
- Depending on the semantics, duplicate records are removed (**set-semantics**) or not (**bag-semantics**).



Unions - Outer union

- When **combining data from multiple sources**, they do not necessarily have compatible schemas.
 - Overlap on some attributes
 - Other attributes are only present in some sources.
- Given two relations $R(A_1A_2, \dots, A_m)$ and $S(B_1B_2, \dots, B_n)$, the **outer union** ($\sqcup$)
 - Creates a relation with schema attributes $\{A_1A_2, \dots, A_m\} \cup \{B_1B_2, \dots, B_n\}$
 - Adds null values ($\perp$) to attributes of R not present in S and vice versa.
 - Unifies the resulting tuples to obtain the final result.



Outer union - example

```
# Sample data for two tables
data1 = {'ID': [1, 2], 'Value1': ['A', 'B']}
data2 = {'ID': [2, 3], 'Value2': ['B', 'C']}

df1 = pd.DataFrame(data1)
df2 = pd.DataFrame(data2)

# (Outer) union of the two tables
union_df = pd.concat([df1, df2]).drop_duplicates()
print(df1)
print(df2)
print(union_df)
```

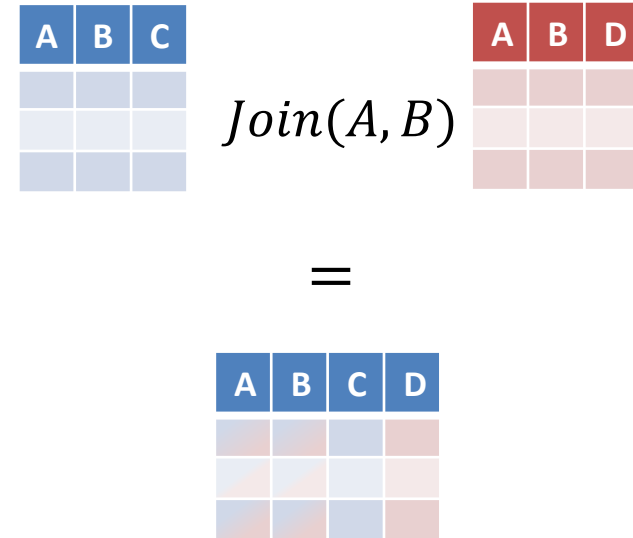
	ID	Value1
0	1	A
1	2	B

	ID	Value2
0	2	B
1	3	C

	ID	Value1	Value2
0	1	A	NaN
1	2	B	NaN
0	2	NaN	B
1	3	NaN	C

Joins

- Linking records from one dataset R to records from another dataset S .
- Linking based on a condition on (join) key attributes, e.g.,
 - Exact match of key values (most common)
 - Approximate match of key values (e.g., to account for data quality issues)



Types of joins for enrichment



Inner join

Only produce a record when there are matching records from each dataset being blended. If there are duplicate join keys, the output records are similarly duplicated.



Left outer join

Retains all records from the left (or initial) dataset, even if there is no matching record in the right (or incoming) dataset. Missing values from the right are padded with null values.



Right outer join

Retains all records from the right (or incoming) dataset, even if there is no matching record in the left (or initial) dataset. Missing values from the left are padded with null values.



Full outer join

Full outer joins retain all records from both datasets, even if they have a corresponding match. Missing values from either side are padded with nulls.

Left outer join - example

```
# Sample data for two tables
data1 = {'ID': [1, 2], 'Value1': ['A', 'B']}
data2 = {'ID': [1, 3], 'Value2': ['B', 'C']}

df1 = pd.DataFrame(data1)
df2 = pd.DataFrame(data2)

# Left outer join of the two tables
joined_df = pd.merge(df1, df2, on='ID', how='left')
print(df1)
print(df2)
print(joined_df)
```

	ID	Value1
0	1	A
1	2	B

	ID	Value2
0	1	B
1	3	C

	ID	Value1	Value2
0	1	A	B
1	2	B	NaN

Generic derivation of values

- Generic derivations apply to many datasets and rely on general rules or formulas.
- May call external services for enrichment.
- Examples:
 - Enrichment to address the time dimension
 - Enrichment to obtain geography or spatial encodings
 - Enrichment based on analyzing data (e.g., text)
 - Values derived from numeric calculations

Proprietary derivation of values

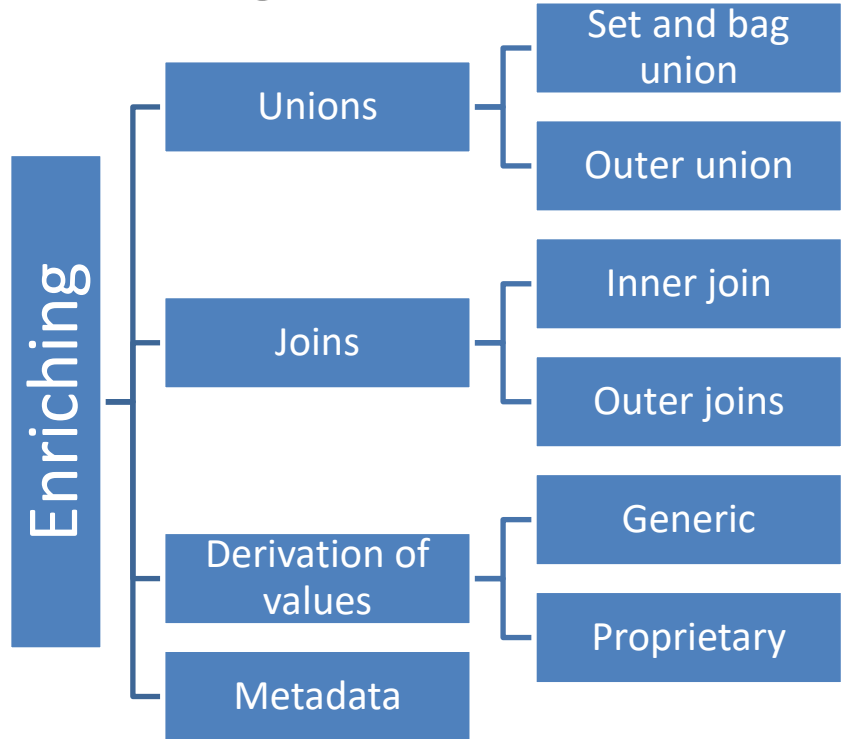
- Organizations may have custom models that derive information to enrich their data.
- Examples:
 - Custom models for credit scoring
 - Custom model for health status of patients
- In data preparation systems, proprietary functionality is often encoded as user-defined functions (UDFs).

Metadata

- (Retrieved) Metadata are integrated into the records as additional attributes.
- Common examples:
 - filenames of the source data
 - Byte offsets and/or record numbers
 - current date and/or time
 - creation/update/access timestamps
 - record and/or record attribute provenance

Summary of enriching

Results in the **addition of information** to a dataset.



DATA CLEANING



Data cleaning overview

- Data cleaning is the process **of identifying and mitigating data errors** within a dataset.
- It is an essential step in data science that ensures that the **quality of data** is high to allow effective and reliable analysis and decision-making.
- **Data cleaning transformation actions** fall in two categories:
 - **Identification** of data quality issues
 - **Correction** or repair of the data

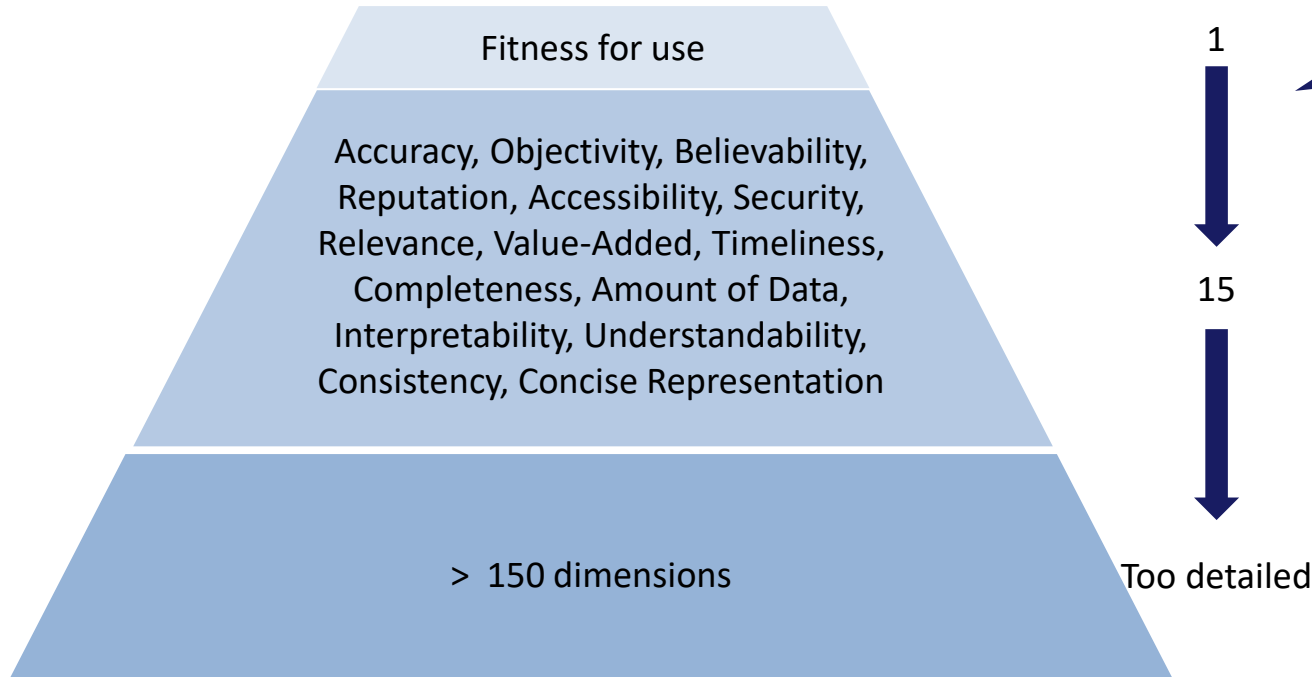
Data cleaning overview

- Throughout this sub-chapter, we cover:
 - **Data quality**: What is it? What dimensions are important?
 - **Data errors**: What types of errors to look out for?
 - **Data cleaning transformation actions**:
Overview of detection and correction transformation actions.

DATA QUALITY



What is data quality?



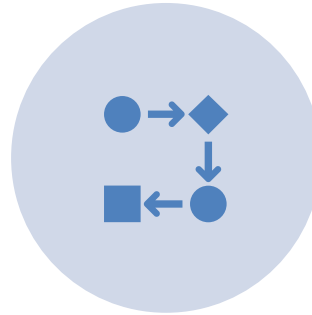
Many dimensions!
Not all of them
fixable by data
cleaning (or
transformation).

Understanding
important
dimensions still
relevant when
developing data
science systems.

Many classifications of quality dimensions



SEMANTIC-ORIENTED CLASSIFICATIONS
PARTITION QUALITY DIMENSIONS SOLELY
BASED ON THEIR MEANING



PROCESS-ORIENTED CLASSIFICATIONS
PARTITION QUALITY DIMENSIONS
ACCORDING TO THEIR DEPLOYMENT IN
DIFFERENT PHASES OF DATA PROCESSING



GOAL-ORIENTED CLASSIFICATIONS
MATCH GOALS THAT ARE TO BE REACHED
WITH THE HELP OF QUALITY REASONING.

Requirement Survey – A comprehensive semantic-oriented classification

Content-related criteria

- Concern the actual data
- Criteria:
 - Accuracy
 - Documentation
 - Relevancy
 - Value-added
 - Completeness
 - Interpretability

Technical criteria

- Concern software and hardware
- Criteria:
 - Timeliness
 - Reliability
 - Latency
 - Performability
 - Response time
 - Security
 - Accessibility
 - Price
 - Customer support

Intellectual criteria

- Concern subjective aspects.
- Criteria:
 - Believability
 - Reputation
 - Objectivity

Instantiation-related criteria

- Concern the presentation of retrieved data.
- Criteria:
 - Verifiability
 - Amount of data
 - Understandability
 - Concise representation
 - Consistent representation

Requirement Survey – Content-based criteria

- **Accuracy** is the extent to which data is correct, reliable, and certified free of error. [WS96]
- **Completeness** is the extent to which data is not missing and is of sufficient breadth, depth, and scope for the task at hand. [WS96]
- **Customer support** is the amount and usefulness of human help via email or telephone.
- **Documentation** is the amount and usefulness of documents with metadata.
- **Interpretability** is the extent to which data is in appropriate languages, symbols, and units, and the definitions are clear. [WS96]
- **Relevancy (or relevance)** is the extent to which data is applicable and helpful for the task at hand. [WS96]
- **Reliability** is the degree to which the user can trust the information
- **Value-Added** is the extent to which data is beneficial and provides advantages from its use. [WS96]

Requirement Survey – Technical criteria

- **Accessibility (or availability)** of a DBMS
 - is the probability that a feasible query is correctly answered must time range.
 - Is the extent to which data are available or easily and quickly receivable [WS96].
- **Latency** is the amount of time from issuing the query until the first data item reaches the user
- **Price (cost effectiveness)**
 - is the amount of money a user must pay for a query.
 - is the extent to which the cost of collecting appropriate data is reasonable [WS96].
- **Response time** measures the delay in seconds between submission of a query by the user and reception of the complete response from the information system.
- **Security** is the extent to which access to data is restricted appropriately to maintain its security [WS96].
- **Timeliness** is the extent to which the age of the data is appropriate for the task at hand [WS96].

Requirement Survey – Intellectual criteria

- **Believability** is the extent to which data is regarded as true, real, and credible [WS96].
- **Objectivity** is the extent to which data is unbiased, unprejudiced, and impartial [WS96].
- **Reputation** is the extent to which data is trusted or highly regarded in terms of its source or content [WS96].

Requirement Survey – Instantiation-related criteria

- **Amount of data** is the extent to which the quantity or volume of available data is appropriate [WS96].
- **Representational conciseness** is the extent to which data is compactly represented without being overwhelming [WS96].
- **Representational consistency** is the extent to which data is always represented in the same format and are compatible with previous data [WS96].
- **Understandability (ease of understanding)** is the extent to which data are clear without ambiguity and easily comprehended [WS96].
- **Verifiability (traceability, lineage)** Is the extent to which data are well documented, verifiable, and easily attributed to a source [WS96].

Requirement Survey – Summary

Content-related criteria

- Concern the actual data
- Criteria:
 - Accuracy
 - Documentation
 - Relevancy
 - Value-added
 - Completeness
 - Interpretability

Multiple content-related criteria can be improved through proper data preparation, in particular data cleaning.

Technical criteria

- Concern software and hardware
- Criteria:
 - Timeliness
 - Reliability
 - Latency
 - Performability
 - Response time
 - Security
 - Accessibility
 - Price
 - Customer support

Technical criteria influence data management and data science deployment considerations

Intellectual criteria

- Concern subjective aspects.
- Criteria:
 - Believability
 - Reputation
 - Objectivity

Intellectual criteria are essential during data selection.

Instantiation-related criteria

- Concern the presentation of retrieved data.
- Criteria:
 - Verifiability
 - Amount of data
 - Understandability
 - Concise representation
 - Consistent representation

Multiple problems related to instantiation-related criteria can be mitigated through data preparation.

DATA ERRORS

Examples of data errors that affect data quality

representation

contradiction

referential integrity

CNr	Name	Birthday	Age	Sex	Phone	ZIP
1234	Smith, John	18.2.80	21	m	999-9999	70567
1234	Jane Doe	32.2.70	46	f	768-4511	55555
1235	John Smith	18.2.80	36	m	567-3211	70567

uniqueness

dummy value

incorrect values

typo

ZIP	Place
70567	Stuttgart
70567	Stutgart
70569	Germany

duplicates

Classification of errors [DR00]

	Single data source	Integrated (clean) data sources
Schema level	Lack of integrity constraints, poor schema design - Invalid value - Violation of attribute dependency - Uniqueness violation - Violation of referential integrity	Heterogeneous data models and schema designs - Labeling conflicts - Modeling conflicts
Instance level	Data entry errors - Missing value - Typographical error or wrong value - Cryptic value - Embedded value - Misfielded value - Contradictory values - Transposition - Duplicates - Data conflicts - Wrong references	Overlapping, contradicting and inconsistent data - Contradictory values* - Different representations - Different scales - Different precision - Different level of aggregation - Duplicates* - Data conflicts*

* Same error types as in a single data source, but the process of data integration can result in additional instances of these error types (even if individual sources are clean)

Single data source – Schema level

Problem		Dirty Data	Reasons/Remarks
Attribute level	Illegal values	bdate=30.13.70	values outside of domain range
Intra-record level	Violated attribute dependencies	age=22, bdate=12.02.70	age = (current date – birth date) should hold
Inter-record level	Uniqueness violation	emp1=(name="John Smith", FIN="123456") emp2=(name="Peter Miller", FIN="123456")	uniqueness of FIN (Foreigner Identification number) is violated
Inter-table level	Referential integrity violation	emp=(name="John Smith", deptno=127)	referenced department (127) is not defined in department table

Single data source – Instance level

Problem		Dirty Data	Reasons/Remarks
Attribute level	Missing values	phone=9999-999999	unavailable values during data entry(dummy values or null)
	Misspellings	city="Bellin"	usually typos, phonetic errors
	Cryptic values, abbreviations	experience="B"; occupation="DB Prog."	
	Embedded values	name="J. Smith 12.02.70 New York"	multiple values entered in one attribute (e.g. in a free-form field)
	Misfielded values	city="Germany"	
Intra-record level	Violated attribute dependencies	city="Redmond", zip=77777	city and zip code should correspond

Single data source – Instance level (continued)

Problem		Dirty Data	Reasons/Remarks
Inter-record level	Word transpositions	name1= "J. Smith", name2="Miller P."	usually in a free-form field
	Duplicated records	emp1=(name="John Smith",...); emp2=(name="J. Smith",...)	same employee represented twice due to some data entry errors
	Contradicting records	emp1=(name="John Smith",bdate=12.02.70); emp2=(name="John Smith", bdate=12.12.70)	the same real-world entity is described by different values
Inter-table level	Wrong references	emp=(name="John Smith", deptno=17)	referenced department (17) is defined but wrong

Integrated data sources – Schema level

Table-table conflicts

- Labeling conflicts
 - Semantically equivalent tables have different names (synonyms)
 - Semantically different tables have equal names (homonym)
- Modeling conflicts
 - Missing attributes
 - Missing but derivable attributes
 - IC conflicts (IC = integrity constraint)

IC conflict
(uniqueness)

Employee

<u>EID</u>	Firstname	lastname	role
1	Peter	Müller	Assistant
5	Petra	Schmidt	Asst.

Employee (Semantic: managing)

Homonym

EID	firstname	lastname
2	Stefanie	Meier
2	Petra	Schmidt

Missing but
derivable
attribute

Integrated data sources – Schema level (continued)

Attribute-attribute conflicts

- Labeling conflicts
 - Different names for semantically equal attributes (synonyms)
 - Same label for attributes with different semantics (homonyms)
- Modeling conflicts
 - Default value conflicts
 - IC conflicts
 - Data type conflicts
 - Condition conflicts

Employee

p_id	firstn VARCHAR(35)	lastname	age
1	Wolfgang	Meyer	33
5	Klaus	Schmidt	NULL
...	...	...	...

IC conflict
(age > 18)

Employee

p_id	firstn VARCHAR(20)	name	age
1	Peter	Müller	0
5	Petra	Weger	17
...	...	...	...

Default
value

Data type

Synonym

Integrated data sources – Instance level

Errors that arise due to the integration of overlapping, contradicting and inconsistent data across sources.

- Contradictory values
- Different representations
- Different scales
- Different precision
- Duplicates
- Data conflicts
- Different level of aggregation

CID	Name	Size	Billing	Phone
1	Smith, John	8.5	80 USD	(123) 4567
2	Doe, Jane	7.0	70 USD	(408) 9012
A	John Smith	42	580 SGD	4444 8888
B	Chris Lim	42	420 SGD	9182 7374

 Data from source 1

 Data from source 2

Data errors summary

- Various types of errors need to be considered.
 - This entails a high variety of solutions to detect and correct these errors.
- Errors are either defined at schema or data level.
 - Knowledge of this distinction helps in choosing suited methods for error detection and correction.

CLEANING TRANSFORMATION ACTIONS

Cleaning actions overview



ERROR DETECTION
IDENTIFIES ERRONEOUS RECORDS
OR ATTRIBUTE VALUES.



ERROR CORRECTION
CHANGES VALUES TO CORRECT
ERRORS.



Error detection

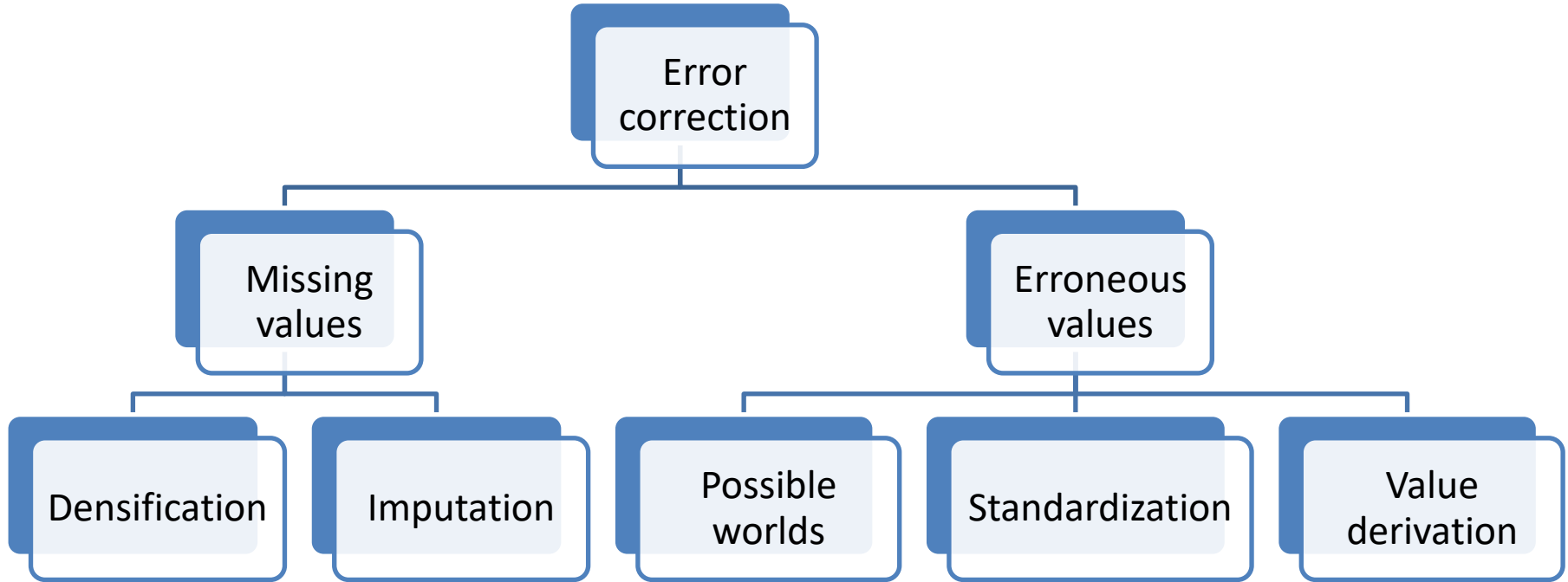
- Can be performed by leveraging metadata gathered during data profiling.
Examples: approximate functional dependencies, cardinalities
- Further data mining or data analysis methods can be used.
Examples: outlier detection, entity resolution to identify different representations of a same entity
- Can produce additional metadata pinpointing detected quality issues, subsequently used to fix them manually, semi-automatically, or fully automatically.
Example: additional duplicate entityID as a result of entity resolution, Boolean flag indicating an error in a specific field.

Error correction



- Distinction between cleaning actions addressing **missing values** and **erroneous values**.
- A simple cleaning action to correct both types of errors is to **delete** (filter) affected records.
- Error **correction beyond deletion** discussed based on the **classification** on the next slide.

Error correction beyond deletion



Correcting missing values

- **Data densification:** merge records that complement each other to reduce / eliminate missing or NULL values.
- **Data imputation:** fills in the missing values through
 - Computation, approximation, or extrapolation
 - Inference based on similar or related records
 - Generation using generative models

Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik's cube	10	NULL
01/01/2024	Bath towel	13	Home
01/01/2024	Rubik's cube	NULL	Puzzle
02/01/2024	Beach towel	NULL	NULL
03/01/2024	Large towel	15	Home
NULL	Jigsaw	20	Puzzle
05/01/2024	NULL	14	Home

Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik's cube	10	NULL
01/01/2024	Bath towel	13	Home
01/01/2024	Rubik's cube	NULL	Puzzle
02/01/2024	Beach towel	NULL	NULL
03/01/2024	Large towel	15	Home
NULL	Jigsaw	20	Puzzle
05/01/2024	NULL	14	Home

Records complement each other and can be densified into one single tuple with no missing values

Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik's cube	10	<u>Puzzle</u>
01/01/2024	Bath towel	13	Home
02/01/2024	Beach towel	NULL	NULL
03/01/2024	Large towel	15	Home
NULL	Jigsaw	20	Puzzle
05/01/2024	NULL	14	Home

Price is approximated based on the price of similar products (e.g., using the average)

Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik's cube	10	Puzzle
01/01/2024	Bath towel	13	Home
02/01/2024	Beach towel	<u>14</u>	NULL
03/01/2024	Large towel	15	Home
NULL	Jigsaw	20	Puzzle
05/01/2024	NULL	14	Home

Category can be inferred by the category of similar products.

Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik's cube	10	Puzzle
01/01/2024	Bath towel	13	Home
02/01/2024	Beach towel	<u>14</u>	<u>Home</u>
03/01/2024	Large towel	15	Home
NULL	Jigsaw	20	Puzzle
05/01/2024	NULL	14	Home

Strong order of records
allows to extrapolate
possible date

Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik's cube	10	Puzzle
01/01/2024	Bath towel	13	Home
02/01/2024	Beach towel	<u>14</u>	<u>Home</u>
03/01/2024	Large towel	15	Home
<u>04/01/2024</u>	Jigsaw	20	Puzzle
05/01/2024	NULL	19	Home

Generative model can generate a likely value

Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik's cube	10	Puzzle
01/01/2024	Bath towel	13	Home
02/01/2024	Beach towel	<u>14</u>	<u>HOME</u>
03/01/2024	Large towel	15	Home
<u>04/01/2024</u>	Jigsaw	20	Puzzle
05/01/2024	<u>XL towel</u>	19	Home



No more missing values

Correcting erroneous values

- **Possible worlds:** Conduct downstream processing (e.g., analysis) both with invalid value and without to assess its impact on possible insights.
- **Standardization:** Uses dictionaries, external services, rules to correct errors.
- **Value derivation:** Obtain correct / consistent value for an attribute that overwrites the original (invalid) value through conversion, calculation, statistical modelling, etc.

Correcting erroneous values

Object	Mass	Type	Surface
Earth	597.20	Terrestrial planet	Solid
Venus	486.70	Terrestril planet	Solid
Jupiter	1.9	Jovian planet	Gas
Uranus	8700	Jovian planet	Gas
Zerez	0.09	Dwarf planet	Ice
Pluto	1.30	Dwarf planet	Glace

Standardization based on fixed library of valid names of solar system object. Replaced by most similar one.

Correcting erroneous values

Object	Mass	Type	Surface
Earth	597.20	Terrestrial planet	Solid
Venus	486.70	Terrestril planet	Solid
Jupiter	1.9	Jovian planet	Gas
Uranus	8700	Jovian planet	Gas
<u>Ceres</u>	0.09	Dwarf planet	Ice
Pluto	1.30	Dwarf planet	Glance

Standardization using domain knowledge mapping planet names to their mass.

Correcting erroneous values

Object	Mass	Type	Surface
Earth	597.20	Terrestrial planet	Solid
Venus	486.70	Terrestril planet	Solid
Jupiter	<u>189870.00</u>	Jovian planet	Gas
Uranus	8700	Jovian planet	Gas
<u>Ceres</u>	0.09	Dwarf planet	Ice
Pluto	1.30	Dwarf planet	Glace

Derivation of correctly formatted value (all others have two decimals).

Correcting erroneous values

Object	Mass	Type	Surface
Earth	597.20	Terrestrial planet	Solid
Venus	486.70	Terrestril planet	Solid
Jupiter	<u>189870.00</u>	Jovian planet	Gas
Uranus	<u>8700.00</u>	Jovian planet	Gas
<u>Ceres</u>	0.09	Dwarf planet	Ice
Pluto	1.30	Dwarf planet	Glace

Correction of typographical error by replacing this outlier value by more frequent type (assume a larger table with few errors)

Correcting erroneous values

Object	Mass	Type	Surface
Earth	597.20	Terrestrial planet	Solid
Venus	486.70	<u>Terrestrial planet</u>	Solid
Jupiter	<u>189870.00</u>	Jovian planet	Gas
Uranus	<u>8700.00</u>	Jovian planet	Gas
<u>Ceres</u>	0.09	Dwarf planet	Ice
Pluto	1.30	Dwarf planet	Glance

Discovered functional dependency
Type → Surface
and value distribution
allow to determine Ice
as the correct value
(assume a more complete relation)

Correcting erroneous values

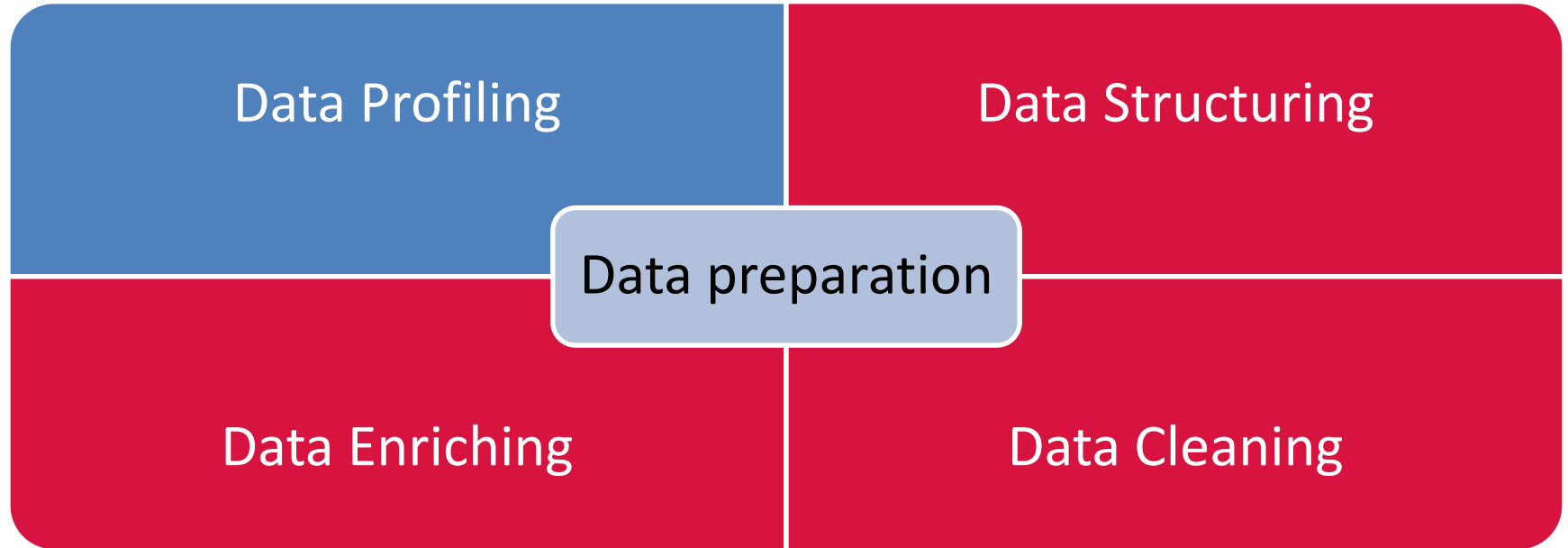
Object	Mass	Type	Surface
Earth	597.20	Terrestrial planet	Solid
Venus	486.70	<u>Terrestrial planet</u>	Solid
Jupiter	<u>189870.00</u>	Jovian planet	Gas
Uranus	<u>8700.00</u>	Jovian planet	Gas
<u>Ceres</u>	0.09	Dwarf planet	Ice
Pluto	1.30	Dwarf planet	<u>Ice</u>



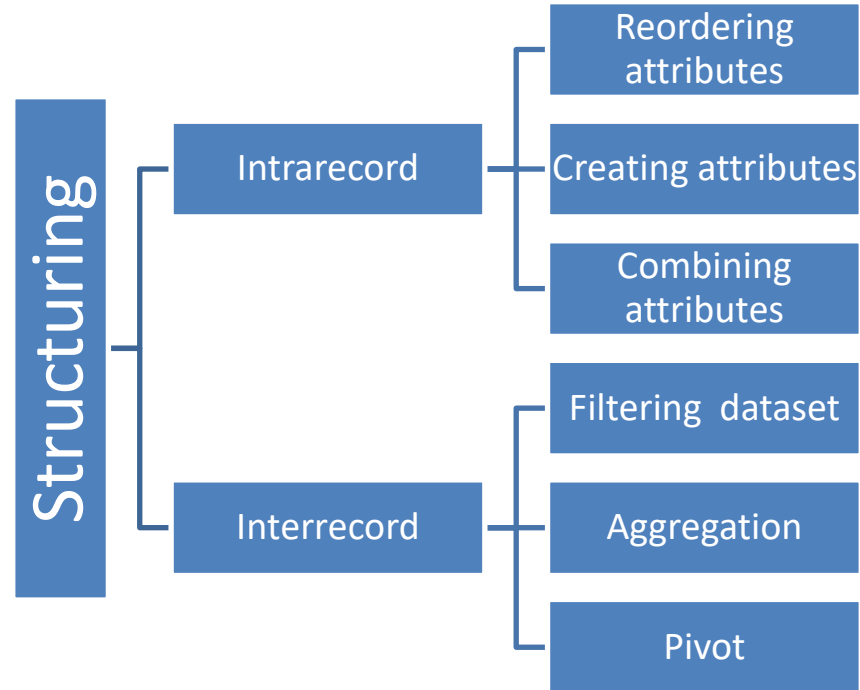
No more erroneous values

DATA TRANSFORMATION CONCLUSION

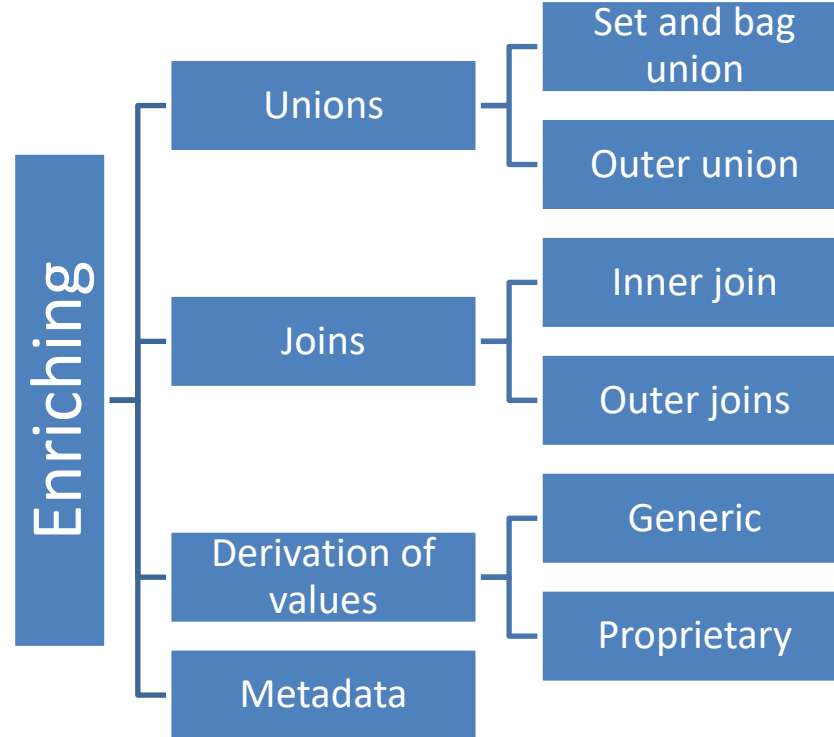
Data transformation tasks



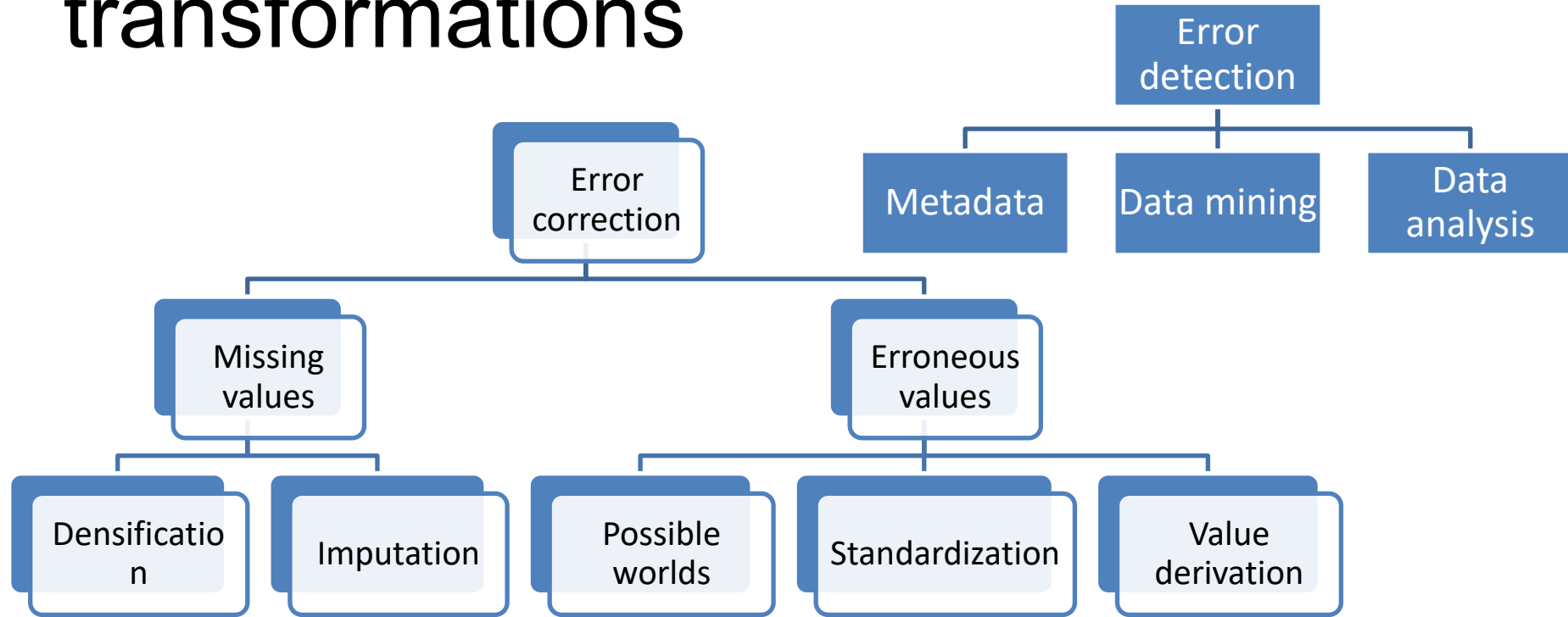
Structuring transformations



Enriching transformations

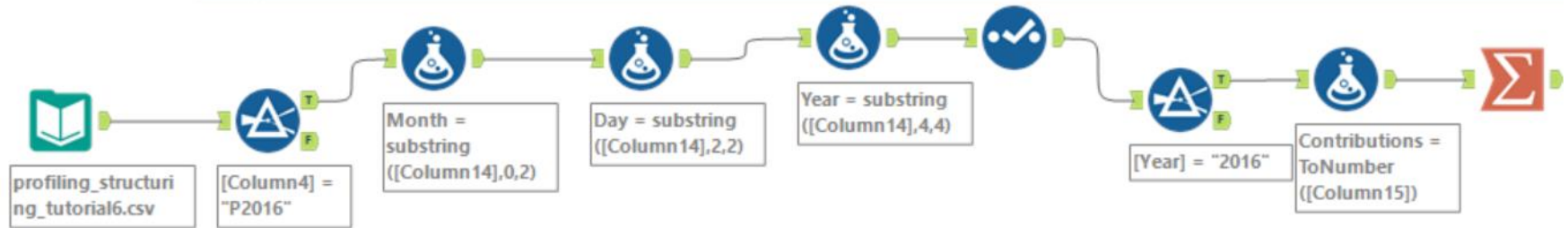
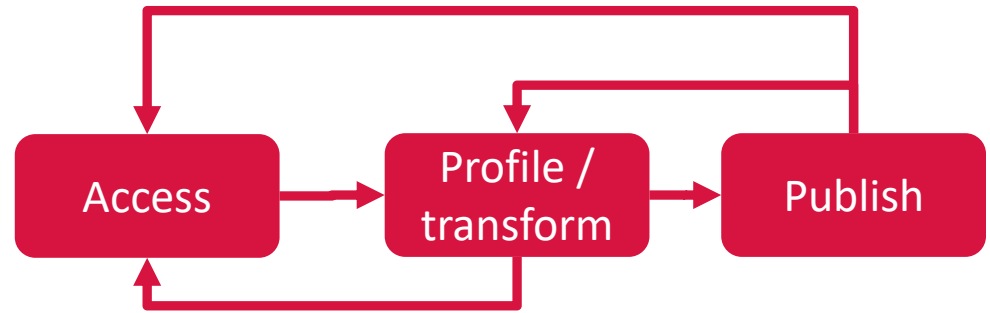


Error detection and correction transformations



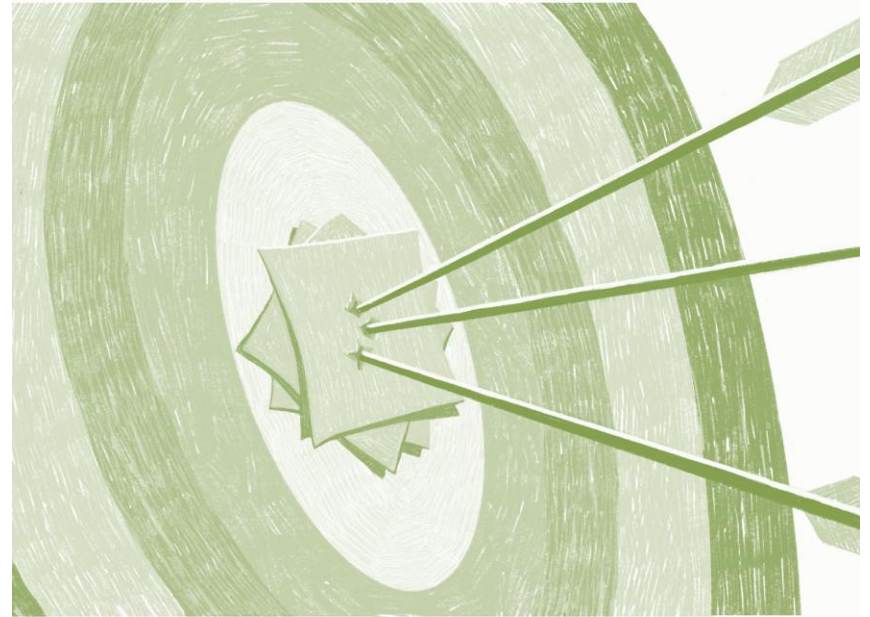
Data preparation is a process

Data preparation is the process of **converting** raw data into a **form** that allows to extract **value** from data.



For each of these types of data transformations, you should now be able to...

- ... define and classify various sub-tasks.
- ... determine which data transformations can be used to prepare the data for a specific data science application.
- ... describe and recognize various data quality issues within and across datasets.
- ... implement various data transformation tasks.



References and credits

- Joseph M. Hellerstein, Tye Rattenbury, Jeffrey Heer, Sean Kandel, Connor Carreras. Principles of Data Wrangling. O'Reilly Media. 2017 – Chapter 4 - 7
- Erhard Rahm, Hong Hai Do: [Data Cleaning: Problems and Current Approaches](#). IEEE Data Eng. Bull. 23(4): 3-13 (2000)
- Felix Naumann, Claudia Rolker: [Do Metadata Models meet IQ Requirements?](#) IQ 1999: 99-114